TITLE PAGE

- **Problem Statement Title -** AI-Driven Yield Optimization Platform for Oilseed Crops
- **Theme -** Agriculture, FoodTech & Rural Development
- **PS Category -** Software
- **Team Name - VisionX**



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Krishi360: Ai-Driven Precision Agriculture Platform For Yield Optimization And Sustainable Farming

Core Concept

The AI-Driven Yield Optimization Platform is a data-intelligent agricultural ecosystem designed to assist oilseed farmers in improving yield, efficiency, and sustainability. It leverages Artificial Intelligence (AI), Machine Learning (ML), Remote Sensing, and IoT-based sensing to generate personalized, data-driven advisories for farmers — covering crop selection, irrigation, fertilizer use, and pest management.

The Problem It Solves

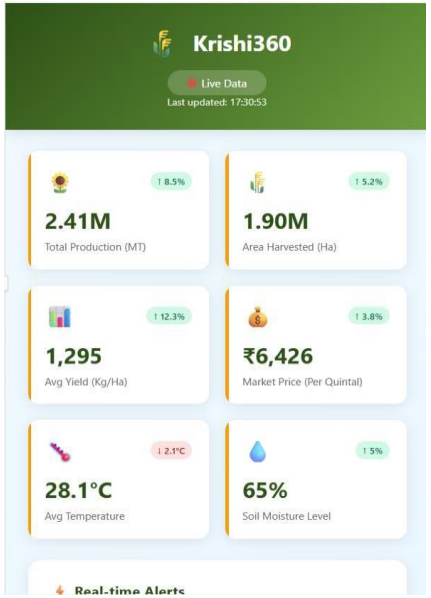
- Traditional farming practices often fail to consider the dynamic interaction between soil nutrients, local weather, and crop physiology.
- Farmers struggle with:
  - Poor awareness of which oilseed suits their soiland
  - Limited ability to interpret soil and weather data.
  - Late identification of diseases, nutrient stress, or pest outbreaks.
  - Lack of personalized recommendations in theirlocal language.
- The platform addresses these issues through predictive analytics and decision automation.

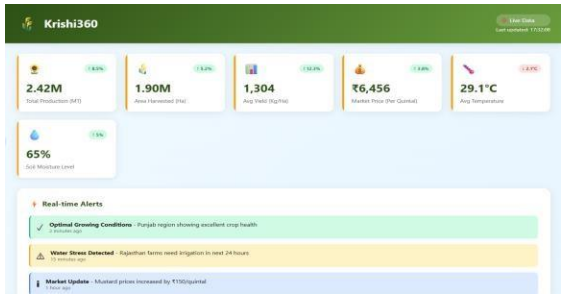


Uniqueness of the Solution

- Localized AI models trained on Indian agro-climatic zones.
- End-to-end ecosystem integrates IoT, satellite, AI, and farmer apps.
- Adaptive learning models evolve with every season's data.
- Multilingual and voice-enabled ensures inclusivity for all farmers.
- Scalable and modular easily extendable to pulses, cereals, or horticultural crops.







Prototype

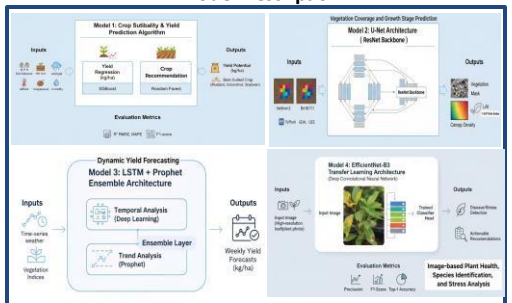
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TECHNICAL APPROACH

1. **Data Collection & Integration :** Aggregates soil sensors, weather APIs, satellite imagery, farmer inputs, and historical yield data. Data is cleaned, validated, and stored in a cloud data lake & feature store.
2. **Four Specialized Ai models for:**
   - Crop & Yield Potential Prediction (Model 1)
   - Vegetation Coverage & Growth Stage (Model 2)
   - Dynamic Yield Forecasting (Model 3)
   - Image-based Health & Stress Analysis (Model 4)
3. **Precision Advisory Engine:**
   - Personalized recommendations based on a farmer's field conditions and crop stage.
   - Localized alerts for irrigation need, pest risk, or nutrient deficiency.
   - Dynamic decision support, updating advisories as new IoT or satellite data arrives.
4. **Visualization & simulation module**:
   - Monte Carlo Simulations: Projects yield outcomes under varying inputs like rainfall, irrigation, fertilizer, and soil fertility.
   - Scenario Testing: Assesses impact of different practices on yield and risk.
   - Interactive Dashboards: Visualizes yield curves and probabilities to support data-driven farming decisions.
5. **Continuous Learning Pipeline: Farmer feedback + IoT data retrain models via MLflow**, ensuring adaptability and accuracy.
6. **System Integration & Deployment:**
   - Architecture: Microservice backend (FastAPI) with ML APIs. Storage: Time-series & imagery on GCS; metadata in PostgreSQL/PostGIS.
   - Frontend: Mobile (Flutter) & web (React) with multilingual support.
   - Alerts: Real-time FCM for crop stress, rainfall, and pest events.

Outcomes: Enables farmers to make informed, data-driven decisions by optimizing interventions, assessing risks, and visualizing yield scenarios.

Ai Model Description



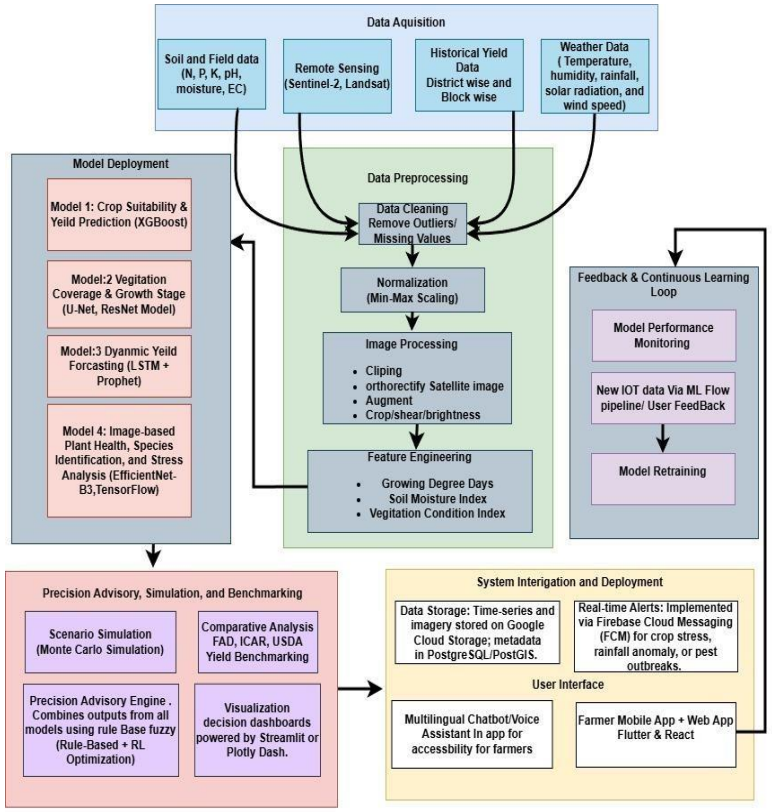
(Model 1) Crop & Yield Potential Prediction : Recommends the best oilseed crop and estimates initial yield based on **soil and climate data** (NPK, pH, moisture, temperature, rainfall).

(Model2) Vegetation Coverage & Growth Stage : Identifies crop coverage, growth stage, and stress areas using **satellite imagery and vegetation indices** (NDVI, EVI, SAVI).

(Model 3) Dynamic Yield Forecasting : Offers **weekly, real-time yield predictions** throughout the season using a **Hybrid LSTM + Prophet** model based on weather and NDVI data.

(Model4) Image-based Health & Stress Analysis : Diagnoses **nutrient, pest, and disease stress** from on-field images using **EfficientNet-B3 CNN** for immediate corrective action.

System Architecture



TECH STACK:-



Programming language: Python, JavaScript, TypeScript, Mongodb, Bash/Shell.
ML/AI: Tensor Flow, Py torch, Scikit-Learn, XG Boost, ML flow.
Database: Post-SQL, Post-GIS, Google Big Query, Firebase RealTime Database.
Google cloud: Google Cloud Storage, Google Compute Engine, Google Kubernetes Engine, Big Query.
Frontend: React, Flutter, Tailwind CSS, Ploty.js/D3.Js, Progressive Web Apps.
IoT / Hardware : MQTT Protocol, Soil/Humidity Sensors, Raspberry Pi/ESP32

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FEASIBILITY AND VIABILITY

❖ FEASIBILITY

Technical Feasibility

- **Reliable data sources**: ICAR, IMD, Sentinel-2, Landsat, IoT sensors.
- **Proven ML/DL models:** XGBoost, Random Forest, U-Net, LSTM-Prophet, EfficientNet.
- Scalable microservice backend (FastAPI + REST APIs); cross-platform UI (Flutter, React).
- Real-time alert system via Firebase Cloud Messaging.

Operational Feasibility

- Automated preprocessing ensures data quality and consistency.
- Integrated simulation & benchmarking modules for transparency.
- Cloud-based architecture supports continuous retraining and nationwide scaling.

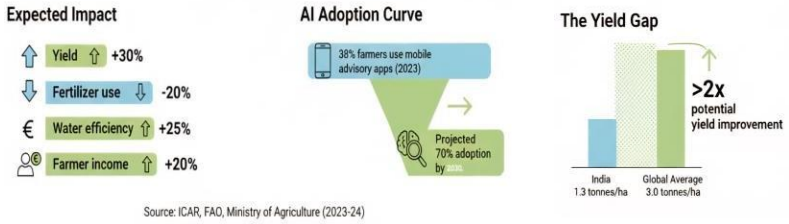
❖ VIABILITY

Economic Viability

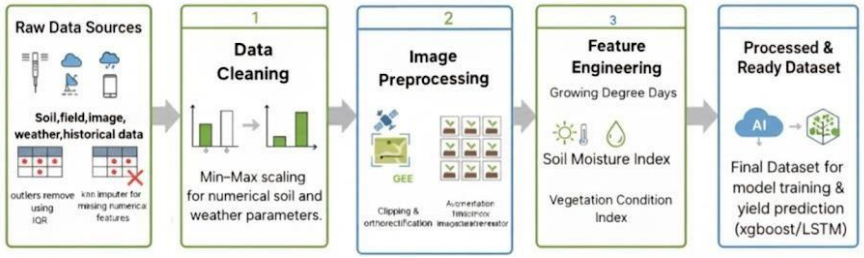
- Reduces edible oil import dependency **(~56%)**, saving up to **1.328 trillion rupees/year**.
- Boosts farmer income by **₹8K–₹12K per hectare** via yield and input optimization.
- Cost-effective deployment using open-source and cloud technologies.

Social & Environmental Viability

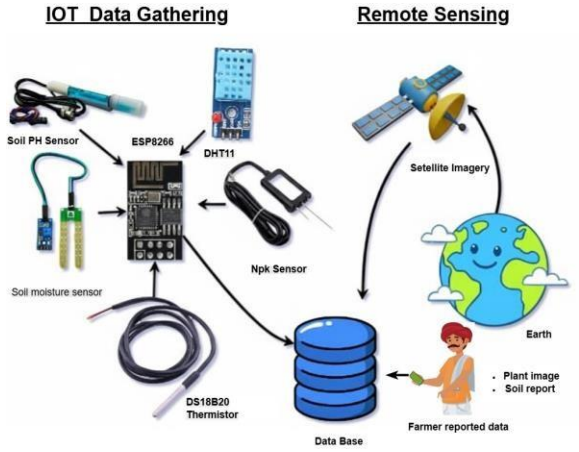
- Empowers smallholders with personalized, multilingual advisories.
- Promotes precision farming — reduced chemical runoff, improved soil health.
- Aligns with **Atmanirbhar Bharat** and **UN SDGs (2, 12, 13)** for sustainable agriculture.



Data Processing pipeline



Data Collection





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IMPACT AND BENEFITS

IMPACTS:-

- **20–30% yield improvement** through AI-driven advisories and precision inputs.
- **25–40% reduction** in water and fertilizer wastage via smart recommendations.
- **Stabilized oilseed acreage** through real-time crop monitoring and alerts.
- **$1.6B potential annual savings** by reducing edible oil imports.
- **Boosts national self-reliance (Atmanirbhar Bharat)** and food security.

FUTURE SCOPE:-

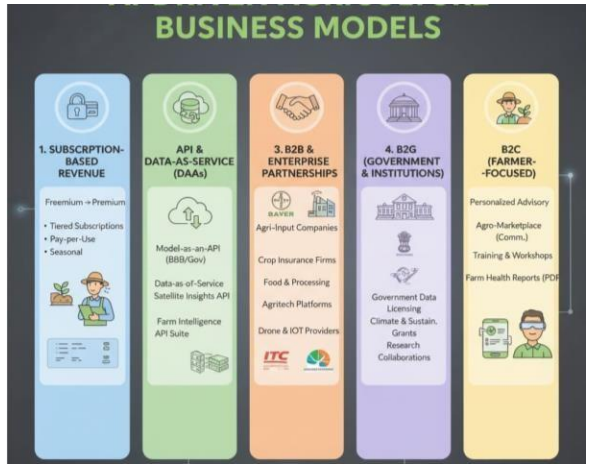
- **Integration** with Digital Agriculture Mission & crop insurance platforms.
- **Partnerships** with ICAR, ISRO, and AgriTech startups for data expansion.
- **Edge AI & drone integration** for on-field analytics.
- **Generative AI chatbots** for 24×7 farmer advisory support.
- **Expansion** to other crops (pulses, cereals, horticulture).
- **Policy dashboards** for real-time national yield insights.

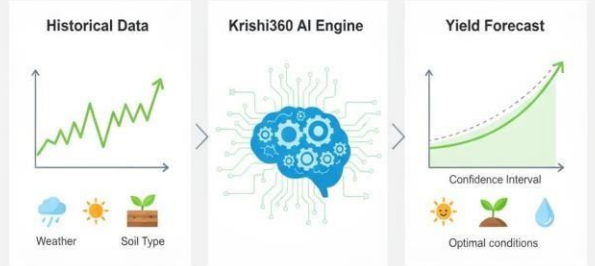
BENEFITS:-

- **Higher farmer income:** ₹8K–₹12K gain per hectare through optimized resource use.
- **Personalized, multilingual support:** Localized advisories accessible via mobile & web.
- **Sustainable farming:** Reduced chemical runoff and improved soil health.
- **Data-driven policymaking:** Benchmarking with FAO/ICAR for better agri decisions.
- **Inclusive access:** Designed for smallholders in rainfed and remote regions.

CHALLENGES & MITIGATION:-

- **Data Gaps:** Partner with ICAR/KVKs; apply cleaning & calibration.
- **Poor Connectivity:** Provide offline mode and SMS/IVR alerts.
- **Low Adoption:** Run multilingual training & demo programs.
- **Regional Variations:** Use federated learning for local adaptation.
- **Data Delays:** Combine multiple APIs (IMD, NOAA, GEE).
- **Privacy Risks:** Ensure encryption & anonymized data storage.
- **Scalability:** Automate retraining via MLflow; deploy on cloud.





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RESEARCH AND REFERENCES

Agriculture & Soil Data Sources

- **ICAR (Indian Council of Agricultural Research):** National agricultural R&D and crop innovation: https://icar.org.in
- **Soil Health Card Portal:** District-wise soil nutrient and fertility data: https://soilhealth.dac.gov.in
- **NABARD:** Agricultural finance and sustainability reports: https://www.nabard.org

Weather & Climate Intelligence

- **NOAA:** Global climate and seasonal pattern datasets: https://www.ncei.noaa.gov
- **NASA POWER:** Agro-climatic, radiation, and environmental parameters: https://power.larc.nasa.gov

Remote Sensing & Satellite Data

- **ISRO Bhuvan:** India-based satellite land and crop data portal: https://bhuvan.nrsc.gov.in
- **Google Earth Engine:** Large-scale geospatial and satellite data analytics: https://earthengine.google.com

Crop & Plant Datasets

- **Plantvillage:** Plant health and disease imagery repository: https://plantvillage.psu.edu
- **Kaggle – Oilseed Data:** Image and yield datasets for oilseed crops: https://www.kaggle.com/datasets
- **FAO FAOSTAT:** Global agricultural production and trade data: https://www.fao.org/faostat

Research Publications

- **PMC Journal:** Modern data-driven agriculture in India: Read Here
- **EOS – Sentinel-2 Satellite Data:** Open access crop and soil monitoring imagery: Read Here

Oilseed Crop	Scientific Name	Suitable Climate / Temperature (°C)	Rainfall Requirement (mm/year)	Soil Type & PH	Soil Moisture/ Drainage	Sunlight & Season	Optimal Growing Regions (India)	Special Agronomic Notes
Mustard	Brassica juncea	10–25°C (cool season)	350–700	Fertile loam or clay loam, pH 6.0–8.0	Moderate; tolerates light frost	Rabi season; high sunlight needed	Rabi season; high sunlight needed	Sensitive to high temp (>30°C) at flowering; sulfur crucial for oil content.
Soybean	Glycine max	20–30°C (humid subtropical)	600–800	Medium black / alluvial soils, pH 6.0–7.5	Moderate moisture; avoid drought	Long day; Kharif crop	Madhya Pradesh, Maharashtra, Rajasthan Madhya Pradesh, Maharashtra, Rajasthan	void water stagnation ; Rhizobium inoculation enhances yield.
Sunflower	Helianthus annuus	20–30°C	500-700	Deep loamy / sandy loam, pH 6.0–7.5	Moderate, well-drained	Needs 6–8 hrs sunlight	Karnataka , AP, Telangana , Maharashtra	Drought- tolerant; nitrogen management affects oil %; avoid saline soils.

Oilseed Crops and Their Optimal Growing Conditions